Creating SELF-EVIDENT DESIGN

Stay tuned for the next exciting chapter in our journey towards design excellence! As you continue through this series, axiomatic design would inspire you to think differently about product development

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So far, we've discussed many important ideas, such as gathering customer and stakeholder needs, analysing risks and costs, and working on the product's appearance. You might be wondering how to take all this information and actually create a product. In this article, we will explore the design thinking process and a helpful tool to turn that information into a well-developed design.

Galileo said in his well-known book, "Once you find them, all truths are easy to understand; the real challenge lies in the process of discovering them." In simpler terms, product designing is like turning different needs into a clear idea that everyone can easily grasp once it's brought to life.

Axiomatic design is a helpful tool for designing things. The name comes from the Greek word 'axíōma,' which means something worthy, fit, or evident. In this process, the design is based on what the customer needs. Axiomatic Design has two key rules:

1. Independence. Corresponding elements from domain to domain should be independent.

2. Information axiom. If two systems are otherwise functionally equal, the system that contains the least amount of information is superior.

In the axiomatic design process, we use a method similar to the QFD analysis we discussed earlier. There are four different domains involved:

1. Customer domain. This is where we focus on understanding the needs and requirements of the customers.

2. Functional domain. Here, we work on defining the functions or tasks that our product or system should perform to meet the customer's needs.

3. Physical domain. This domain deals with the physical elements and components that will be used to realise the functions identified in the functional domain.

4. Process domain. In this domain, we consider the processes and methods needed to create and manufacture the product or system.

These domains help us systematically design and develop products that align with customer needs and perform well. (See Fig. 1 for a visual representation.)

In the customer domain, we collect all the things the product needs to do for the customers. In the functional domain, we check if these ideas are practical. We consider technical and economic limitations and determine what we want to achieve. Then, in the physical design domain, we figure out how to fit all the parts together to make it work as planned. Once we have the design set, we move on to the process domain, where we think about how to actually make the product.

The axiomatic design uses information from QFD and applies two rules to make the design simpler and

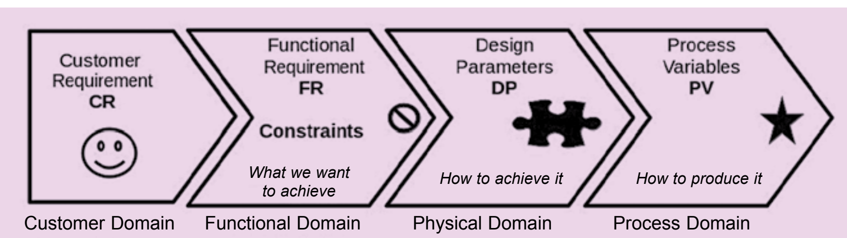

Fig. 1: Four domains of axiomatic design